Mahmoud Alyosify

Edge & Efficient AI Engineer | Inference Optimisation, Latency & Energy-Aware Deployment
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PROFESSIONAL SUMMARY

Efficient-AI Engineer whose entire focus is the constraint most ML work ignores: **making models cheap enough to actually run**. Master's research at **Queen's University, Canada** (expected **Nov 2026**, **Digilians Presidential Scholarship**) tests a **training-free decoding-time method** for cutting wasted LLM output against a strict truncation baseline, measuring **TTFT, inter-token latency and GPU energy through NVML** — efficiency treated as a measured result, not a claim. Practical deployment experience with **ONNX export and FAISS-backed inference**, **4-bit quantised fine-tuning (QLoRA)**, **self-hosted model serving**, and **deep-RL resource provisioning under cost–latency budgets**. Systems background in **C/C++ and real-time national toll infrastructure**, plus a **200-hour ML internship with ITIDA / Egypt Makes Electronics**; currently an **AI Engineer Intern at Injaz Tech**, running a RAG system on a local 3B model on a 4 GB GPU. Seeking edge, on-device and inference-optimisation engineering roles.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Deep Learning, Cloud Computing, Reinforcement Learning, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — real-time contactless vital-sign estimation from video (rPPG). **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Inference Optimisation	Decoding-Time Token Reduction, Custom LogitsProcessors, Entropy-Based Gating, KV-Cache-Free Methods, Quantised Fine-Tuning (QLoRA / 4-bit), PEFT / LoRA Adapters, Self-Hosted Model Serving
Profiling & Measurement	Time-to-First-Token (TTFT), Inter-Token Latency, Throughput Benchmarking, **GPU Energy Measurement via NVML**, Deterministic Benchmarking on Pinned Hardware, Cost–Latency Trade-off Analysis, Per-Node Token & Cost Accounting
Deployment Runtimes	ONNX, FAISS Vector Search, Ollama (Local LLM Serving), Docker, FastAPI, Response Caching, Streamlit, NVIDIA A40 GPU, AWS, Linux, CI/CD
ML & Deep Learning	PyTorch, TensorFlow, Hugging Face Transformers, CNNs, Self-Supervised & Contrastive Learning, Reinforcement Learning (PPO, DQN), Scikit-learn, Optuna
Systems & Low-Level	C, C++, Real-Time Application Support, File-System Implementation (Mini-FAT), Embedded-Oriented ML Training (ITIDA / Egypt Makes Electronics, 200 hrs), Git/GitHub, Bash
Programming	Python, C, C++, C#, SQL, Go, MATLAB, Bash

SELECTED EFFICIENCY & DEPLOYMENT PROJECTS

MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation — *2026 – Present*
MSc Research Project (CISC 898), Queen's University · PyTorch · NVML · NVIDIA A40 GPU

- Research on **training-free decoding-time control** of LLM verbosity — no retraining, no KV-cache overhead — on **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B**, measuring **TTFT, inter-token latency and GPU energy (NVML)** in deterministic runs on a pinned **NVIDIA A40**. An entropy-gated logits processor cut **25.5–34.9%** of MT-Bench tokens within pre-registered BERTScore margins, but an audit showed the saving was truncation.
- The follow-up **Entailment-Licensed In-Decoding Elision** (sentence encoder + NLI model) beats truncation on every model but removes only 0.9–2.2% of tokens: ordinary answers contain little restatement, and larger cuts discard content.

SimCLR-Vision-SSL — From Trained Encoder to ONNX Retrieval Service — *2026*
PyTorch · SimCLR · SupCon · ONNX · FAISS

- Exported a self-supervised vision encoder to **ONNX** and served its embeddings through a **FAISS** index, turning a research model into a runnable, framework-independent retrieval component.
- Reached **84.30% top-1 accuracy** after a **42-experiment augmentation sweep** plus a semi-supervised **SupCon** extension.

RL-Cloud-Autoscaler — Learned Provisioning Under a Cost–Latency Budget *2026*
PPO · DQN · Custom Gymnasium Environment · Resource Optimisation

- Built a custom simulation environment and trained **PPO** and **DQN** agents to allocate compute under an explicit **cost–latency reward**, benchmarked against static and threshold-based policies.

Horus-OSINT — Quantised Fine-Tuning & Self-Hosted Serving *2026*
Meta-Llama-3-8B · Fine-Tuning · AWS · Self-Hosted Inference

- Fine-tuned **Meta-Llama-3-8B** on **159,826** records and deployed it on **AWS** for self-hosted interactive inference — keeping model weights and data in-environment rather than calling a hosted API.

Additional Projects

- **AudioShield** — compact **CNN** over mel-spectrograms delivering real-time audio classification through a lightweight served application.
- **Vitalism Solution** — real-time video signal processing (rPPG + wavelet filtering) for contactless vital-sign estimation from a standard camera.
- **Also:** Mini-FAT File System (C, low-level systems) · HORUS Sentinel (multi-agent, self-hosted LLM) · SCRAP · Automated Multimodal Agent.

PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Ran a **LangGraph** RAG system on a local **Qwen2.5-3B** (Ollama) on a 4 GB GPU; profiled per-node cost (the relevance grader took **75%**).
- An answer cache cut repeat-question latency from **34.7 s to 1.6 s**.

Part-Time Instructor — Machine Learning & AI for Business *Jul 2025 – Present*
National Telecommunication Institute (NTI) Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported **real-time backend systems** for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, where latency and uptime were operational requirements, not metrics on a dashboard.
- Debugged and optimised live transportation software and supported legacy applications to minimise downtime; co-developed an internal data-management application in **C#, .NET, Entity Framework, SQL Server**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)

Machine Learning Specialization — DeepLearning.AI

Digilians Presidential Scholarship — Queen's University

Semi-Finalist — Microsoft Imagine Cup 2023

AWS Academy — ML Foundations, ML for NLP, Data Engineering

HCIP – AI (Huawei)

1st Place — Smart Cities Hackathon 2022

Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.